1 Figure 1.1 shows a student holding a rubber ball 1.4 m above a hard floor. The mass of the ball is 0.18 kg.

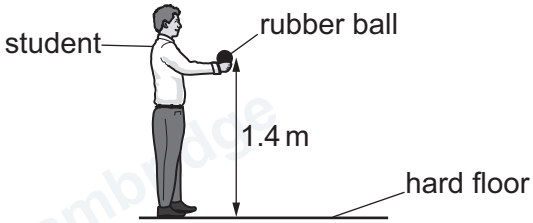


Figure 1.1

(a) State what is meant by the mass of the ball.

.....

..... [1]

(b) The student releases the ball. The ball accelerates downwards due to a force *W*.

(i) State the name of force *W*.

..... [1]

(ii) Calculate *W*. State the equation you use.

W = ... N [2]

(iii) The ball falls 1.4 m. Energy is transferred from the gravitational potential energy store of the ball.

Show that the energy transferred is 2.5 J.

[2]

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(iv) The energy in **1(b)(iii)** is all transferred to the kinetic energy store of the ball.

Calculate the velocity of the ball.

velocity = ... m/s [3]

(c) The ball hits the floor and moves upwards. The magnitude of the velocity is less than the value calculated in **1(b)(iv)**.

State **one** other way the velocity of the ball is different from the velocity calculated in **1(b)(iv)**.

..... [1]

[Total: 10]

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